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Summary

Backend and platform engineer. I lead engineering on codePost, the grading platform Rutgers CS runs for 1,950+ students, graders, and instructors: I rebuilt its autograder, built its AI features and cost tracking, and keep it running with the sysadmin team. I care most about concurrency, isolation, security, and accessibility, and about building what the people using a system actually ask for.

Experience

Rutgers University, Application Developer (Software Engineer)

June 2025 - Present

- Lead engineer on codePost, the grading platform Rutgers CS runs for 1,950+ students, graders, and instructors, which other institutions have asked to self-host. I own the backend, shape what ships by watching how instructors and graders use it, and wrote the docs the team deploys from.
- Drive AI tooling adoption across the team: moved three sysadmins from chat web UIs to CLI agents that gather host data over SSH and draft deployment plans for them, wrote the agent rules both codePost repos share, and mentor a colleague on AI-assisted development.
- Took autograder adoption from three courses to five in a semester by rebuilding it after instructors found the inherited one hard to use; submissions now run in isolated containers across seven languages and Jupyter notebooks.
- Brought codePost's frontend to WCAG 2.1 AA for the ADA Title II deadline and built the monthly scanner that audits the ~100 websites LCSR is responsible for and emails each owner their issues with fixes; many have since been fixed.
- Reworked the grading console with the graders who use it: reusable comment templates after they said feedback on simple assignments was mostly copy-paste, keyboard navigation, inline code execution, and AI-drafted comments.
- Built the platform's AI grading assistance, from comment drafts graders can accept or edit to per-token cost tracking, so every course can see exactly what its AI usage costs and run on its own rates.
- Automated recovery across the platform: autograder images verify, promote, and roll back on their own, unhealthy containers restart themselves, and deploys go out in health-gated stages. Grew the backend test suite from 106 tests to 984.
- Found and fixed a directory traversal in dataset uploads, authorization gaps that let students run code and read unreleased feedback, and an email loop that could exhaust a worker.

Rutgers University, Student Lab Technician

January 2023 - May 2025

- Ran 10 Hackerspace workspaces that drew about 60 students a semester, most of them asking about the AI projects on display; wrote the student-facing documentation and maintained the WordPress site.
- Built two of those demos: a Raspberry Pi computer-vision station on MediaPipe used for walk-in workshops, and an Express and React playground for AI bots.
- Built the LED matrix sign the Hackerspace still runs today, with a web-based pixel and GIF editor and a display schedule.
- Built a 3D printer monitoring service (Express and React) streaming live video of the printer bay so students could check job status without walking over.

Swish, Full-Stack Developer

July 2021 - December 2021

- Built the tooling Swish used to create and manage about 500 accounts across web platforms: Puppeteer automation driven from a Node.js CLI, with Mocha regression tests covering each platform's signup flow.
- Turned the CLI into an Electron desktop app so staff who didn't want a terminal could manage accounts themselves.

C-Tech, Computer Technician

July 2019 - December 2021

- Ran the repair desk: Windows installs, malware removal, and troubleshooting for walk-in customers, plus a secure drive-erasure workflow that supported 3 enterprise clients.
- Automated the Windows build with PowerShell scripts, cutting setup from 40 to 20 minutes per machine.

Skills

Languages: Python, TypeScript, JavaScript, C, Java, SQL, Bash, PowerShell

Frameworks: Django, Django REST Framework, Django Channels, Celery, Flask, Node.js, Express, React, Next.js, JupyterLab, Electron

Data: MariaDB / InnoDB, MongoDB, Redis, Django ORM, Schema Design and Migrations, Query Optimization and Indexing, Caching

Infrastructure: Docker, Docker Compose, Ansible, Nginx, Linux, systemd, Proxmox, VMware, OpenStack, GitHub Actions, GitLab CI, CI/CD, Zabbix, Prometheus, Grafana

Security: Application Security, OWASP Top 10, RBAC / ABAC Authorization, Sandboxing and Container Isolation, CAS / Kerberos / Azure AD SSO, JWT and API Key Management, Suricata, pfSense, Wazuh

Testing: pytest, Vitest, Jest, Playwright, Mocha, axe-core, Molecule, WCAG 2.1 AA

AI: OpenAI and Gemini APIs, Ollama, LLM Cost Metering, Tool-Calling Agents

Projects

CompLock

November 2023 - March 2024

- Built SSH command-and-control software for networked computers in TypeScript on LevelDB and SSH2, used by RUSEC for one CCDC season.
- Rotated passwords across the team's 30 machines in one parallel SSH pass instead of one machine at a time, with Mocha tests running in GitHub Actions.

mohamadk.com

April 2025 - Present

- Designed and built a Next.js 16 / React 19 portfolio and blog with a markdown publishing pipeline, PWA support, and a keyboard command palette.
- Built an in-browser code playground on WebContainers that boots Node.js client-side to run my published ts-declaration-json package live.
- Wrote a simulated Linux shell in TypeScript (filesystem, users, coreutils) that serves as the site's 404 page, plus an AI task-planning agent with tool calling.

Accessibility Scanner

2025 - Present

- Built the monitor LCSR uses for ADA Title II compliance in place of a spreadsheet of sites: a Flask API and Next.js interface drive Playwright and axe-core scans through Celery workers across the ~100 websites the department is responsible for, deployed through an Ansible role I wrote.
- Scans every site monthly and emails the responsible faculty member a report with each issue linked to its fix, since the department tracks the sites but their owners fix them; many have since been fixed.
- Added an endpoint that hands an AI agent a page's issues and the context to fix them, so a site owner can point an agent at their app and the scanner and have it work through the problems.
- Stores per-page results with screenshots so remediation progress is measurable over time.

Education

Bachelor of Arts, Computer Science

May 2025

- Rutgers University. Coursework: Computer Security, Software Methodology, Computer Architecture.

Certifications

AWS Certified Cloud Practitioner, Amazon Web Services

January 2025, valid through January 2028

Leadership

RUSEcurity (Student Cybersecurity Club), Vice President

January 2023 - May 2025

- Managed network administration for the 12-person CCDC team, which placed 4th of 10 teams in 2024.
- Led club infrastructure setup on Proxmox and VMware ESXi, standing up the practice range the team trained on.
- Led development of the club's BlackBox machine environment with Wazuh detection rules and the ToolBox hardening toolkit used for CCDC preparation.